SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO4 – Precision	**Assessment:**	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
Student Name:	M.MANASA	**Reg. No.:**	192425189
Section / Batch:	2024 AIML	**Date:**	27/8/26

Code of Conduct

I __M.MANASA(192425189)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______m.manasa____________________

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Frame Semantics in Banking Customer-Service Systems
A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.
Field Data:

Soil Condition	Irrigation Status
Dry	Available
Wet	Available
Dry	Unavailable
Wet	Unavailable

The system uses the following predicate rules:

- Dry(x) ∧ IrrigationAvailable(x) → NeedsWater(x)
- NeedsWater(x) → Irrigate(x)
- Wet(x) → ¬NeedsWater(x)
- IrrigationUnavailable(x) → ¬Irrigate(x)
- Rice(x) ∧ NeedsWater(x) → PriorityIrrigation(x)

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System
A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company

Search Query	Possible Sense 1	Possible Sense 2
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.
Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing
A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.
The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data

Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	5	5	5	5	5	25	93
2	5	5	5	5	5	25	
3	5	5	4	4	4	22	
4	5	4	4	4	4	21	

Faculty In-charge	Course Coordinator	Program Director
T.KUMARAGURUBARAN		
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q1. Frame Semantics in Banking Customer-Service Systems

Question

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

Answer

•Transfer request → TRANSFER(SourceAccount, Amount, DestinationAccount, Customer). The action is TRANSFER. The participants are the source account, amount, destination account, and customer.

• Debit-card request → BLOCK(DebitCard, Customer). The action is BLOCK and the participants are the debit card and customer.

• Statement request → SEND(TransactionStatement, Email, Customer). The action is SEND and the participants are the transaction statement, destination email, and customer.

• Withdrawal-limit request → MODIFY(WithdrawalLimit, Customer). The action is MODIFY and the participant is the customer's withdrawal limit.

• Incorrect interpretation: B4. The actual intent is "Increase withdrawal limit," while the system interpreted it as "Decrease withdrawal limit." This reverses the intended banking operation.

• Incorrect semantic roles can cause serious errors. For example, confusing source and destination accounts may transfer money to the wrong account, while confusing increase with decrease may reduce a customer's withdrawal limit.

• Recommended approach: use frame-based semantic parsing to identify the action and explicitly label roles such as source, amount, destination, object, customer, and target. Validate important roles against the query and request confirmation when ambiguity could cause a financial error.

Q2. First-Order Predicate Calculus for Smart Agriculture

Question

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

Answer

• F1: Dry(F1), IrrigationAvailable(F1), Rice(F1). F2: Wet(F2), IrrigationAvailable(F2), Wheat(F2). F3: Dry(F3), IrrigationUnavailable(F3), Maize(F3). F4: Wet(F4), IrrigationUnavailable(F4), Rice(F4).

• Given Dry(x) ∧ IrrigationAvailable(x) → NeedsWater(x), F1 satisfies both conditions, so NeedsWater(F1).

• From NeedsWater(x) → Irrigate(x), F1 requires irrigation: Irrigate(F1). Because F1 is rice and needs water, PriorityIrrigation(F1) also follows.

• F2 is wet, so Wet(F2) → ¬NeedsWater(F2). Therefore, F2 does not require irrigation.

• F3 is dry, but irrigation is unavailable. The rule for needing water cannot establish an irrigation action, and IrrigationUnavailable(F3) → ¬Irrigate(F3). Therefore, F3 must not be irrigated.

• F4 is wet, so ¬NeedsWater(F4), and irrigation is unavailable, so ¬Irrigate(F4).

• Final decision: F1 requires irrigation and has priority. F2 and F4 do not require irrigation. F3 is dry but cannot be irrigated because irrigation is unavailable.

• F1 and F3 must be treated differently: F1 is dry with irrigation available, while F3 is dry with irrigation unavailable.

• Predicate logic is effective for definite rule-based decisions, but it is not sufficient by itself when sensor data is incomplete or uncertain. Practical systems need uncertainty handling such as confidence values, probabilistic reasoning, or sensor-fusion methods.

Q3. Word Sense Disambiguation in a Travel Recommendation System

Question

1. Determine the intended sense of the ambiguous terms using the available context.

2. Identify the lexical and contextual cues that distinguish the possible senses.

3. Analyze why the same word may require different interpretations for different users.

4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

Answer

• Amazon tour → Amazon rainforest. “Tour” and the user’s viewing of rainforest trekking packages indicate the travel/geographical sense.

• Safari booking → Wildlife tour. “Booking” and the selection of wildlife resort packages indicate the travel sense rather than the browser.

• Java vacation → Java island. “Vacation” and viewing hotels in Bali and Java indicate the geographical/travel sense rather than the programming language.

• Cruise package → Ship journey. The user viewed Mediterranean ship packages, so “cruise” refers to a ship journey rather than cruise software.

• Safari download for Windows → Web browser. “Download,” “Windows,” and the browser download-page click identify the software sense.

• Useful lexical/contextual cues include words such as tour, booking, vacation, download, and Windows, together with viewed pages, selected packages, previous searches, and clicks.

• The same word can have different meanings for different users because interpretation depends on context and user intent. For example, Safari may mean a wildlife trip to one user and a browser to another.

• Industrial WSD strategy: encode the query and context using embeddings; combine these representations with user-history and click-behaviour features; score candidate senses with a ranking/classification model; use the selected sense for retrieval; and use later clicks/selections as feedback to improve the model.

Q4. Syntax-Driven Semantic Analysis in Legal Document Processing

Question

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

Answer

• In active voice, the grammatical subject commonly performs the action and the object receives it. In passive voice, the grammatical subject may be the affected entity, while the semantic agent may appear in a "by" phrase.

• "The supplier delivered the equipment to the buyer." → Supplier = Agent, Equipment = Object, Buyer = Recipient. This interpretation is correct.

• "The buyer received the equipment from the supplier." → Buyer = Agent/receiver, Equipment = Object/Theme, Supplier = Source. Therefore, the given assignment Supplier = Recipient is incorrect.

• "The company terminated the contract after the violation." → Company = Agent and Contract = Object. This interpretation is correct.

• "The contract was terminated by the company." → Contract = Object/Theme and Company = Agent. The given interpretation reverses these roles, so Contract = Agent and Company = Object are incorrect.

• Confusing grammatical subject with semantic agent can cause a legal NLP system to assign an action to the wrong party. This can produce incorrect extraction of who performed an action, who was affected, and who received it.

• Recommended architecture: dependency/syntactic parsing → voice detection → semantic-role labeling → prepositional-phrase analysis → coreference and long-distance dependency handling → final legal event representation.

• The system should explicitly recognize active/passive constructions and the semantic functions of "by," "from," and "to" phrases before generating the final roles.